

Comprehensive Capstone Project Report

End-to-End Mutual Fund Analytics Pipeline & Presentation Layer

1. Executive Summary

This report provides a comprehensive summary of the **Mutual Fund Analytics Capstone Project**, a production-grade data engineering and quantitative analytics system designed for the Indian Mutual Fund industry.

Core Accomplishments

- Automated ETL Pipeline:** Built an end-to-end pipeline (`data_ingestion.py`, `live_nav_fetch.py`, `clean_and_load.py`) that ingests raw transaction, master, and index datasets, cleans data quality anomalies, handles weekday reindexing with forward-filling for holidays, and loads them into a normalized database.
- Relational Database Model:** Designed and populated a star schema in a SQLite database (`bluestock_mf.db`) containing 10 inter-linked dimension and fact tables, holding **over 90,000 records** of historical data.
- Quantitative Analytics Engine:** Programmed financial calculations for Trailing Returns, Compounded Annual Growth Rate (CAGR), Sharpe Ratio, Sortino Ratio, CAPM Alpha, Beta, and Maximum Drawdown (MDD) across 40 schemes over a 4.5-year history.
- Advanced Risk & Cohort Modeling:** Formulated and calculated daily Value at Risk (VaR), Conditional VaR (CVaR), 90-day rolling Sharpe ratios, Herfindahl-Hirschman Index (HHI) for sector concentrations, and customer cohort retention tracking.
- Interactive Web Dashboard:** Developed a 6-page interactive web interface served via a Python backend server (`dashboard/server.py`) offering filters, charts (via Chart.js), scorecards, drill-through analytics, and a recommendation engine.

2. System Architecture

The pipeline follows a modern decoupled data engineering architecture:



4. Presentation Layer (Dashboard)

- Python Server (dashboard/server.py API)
- HTML5/CSS3/JS Web Interface (Chart.js)

3. Quantitative & Risk Engine

- CAGR, Sharpe, Sortino, CAPM Alpha/Beta
- VaR, CVaR, HHI, Cohorts, SIP Continuation

Components

- **Data Sources:** Historical daily NAVs, investor transactions (32,778 rows), AMC AUM statements, category inflows, industry folios, benchmark index values (8,050 rows).
- **Storage Layer:** Relational SQLite3 database optimizing storage footprint and enabling complex multi-table joins.
- **Backend Layer:** Multi-threaded Python server serving dynamic REST API endpoints for key metrics, recommendations, cohorts, and fund-specific details.
- **Frontend Layer:** Responsive web interface written in vanilla HTML5, CSS3, and ES6 Javascript, relying on Chart.js for visualization and Font Awesome for icons.

3. Database Schema Design (Star Schema)

The analytical database `bluestock_mf.db` implements an optimized Star Schema with strict types, constraints, and index-ready primary and foreign keys:

dim_fund (Fund Master Table - Primary Dimension)

amfi_code (PK) | scheme_name | fund_house | category | sub_category | plan | launch_date | benchmark | expense_ratio_pct | min_sip_amount

▼ 1-to-Many Relationships

fact_nav amfi_code (FK, PK) nav_date (PK) nav daily_return	fact_performance amfi_code (FK, PK) return_3yr_pct (CAGR) sharpe_ratio sortino_ratio alpha beta max_drawdown_pct aum_crore expense_ratio_pct	fact_transactions transaction_id (PK) investor_id transaction_date amfi_code (FK) transaction_type amount_inr state age_group payment_mode
fact_portfolio_holdings amfi_code (FK, PK) stock_symbol (PK) stock_name sector weight_pct market_value_cr	fact_advanced_risk amfi_code (FK, PK) var_95_pct cvar_95_pct	fact_sector_concentration amfi_code (FK, PK) scheme_name hhi_index concentration_level

4. Data Pipeline & ETL Process

Data Quality Auditing & Cleaning

During ingestion, multiple data quality anomalies were identified and resolved programmatically in `clean_and_load.py`:

- **Reindexing to Standard Business Days (`freq='B'`):** The raw `02_nav_history.csv` was missing market holidays. To prevent gaps in calculations, each scheme's NAV series was reindexed to include every Monday-through-Friday business day (1,150 unique dates from Jan 3, 2022 to May 29, 2026). The missing NAV values were forward-filled (`ffill()`).
- **Transaction Standardization:** Cleaned `transaction_type` in `08_investor_transactions.csv` to map strictly to SIP, Lumpsum, OR Redemption. Transaction amounts were verified to be strictly positive (> 0).
- **KYC Field Correction:** Sanitized KYC status flags to ensure values map strictly to `verified` or `Pending`.
- **Out-of-Bounds Ratios:** Audited expense ratios to verify they lie in standard boundaries $[0.1\%, 2.5\%]$. Corrected drawdown representation (standardized as negative values).

5. Quantitative Analytics Calculations

The engine performs critical portfolio risk and return evaluations based on the following formulas:

Trailing NAV Return

Daily Return Formula:

$$\text{Return}_t = (\text{NAV}_t - \text{NAV}_{t-1}) / \text{NAV}_{t-1}$$

Compounded Annual Growth Rate (CAGR)

Annualized growth rate over a fund's historical span:

Compounded Annual Growth Rate (CAGR) Formula:

$$\text{CAGR} = (\text{Ending Value} / \text{Beginning Value})^{(365.25 / \text{Days})} - 1$$

Sharpe Ratio

Risk-adjusted performance utilizing an annualized risk-free rate (R_f) of 6.0%:

Sharpe Ratio Formula:

$$\text{Sharpe Ratio} = (R_{\text{annualized}} - R_f) / \sigma_{\text{annualized}}$$

Where $R_{\text{annualized}} = \text{Mean}(\text{Daily Return}) \times 252$, and $\sigma_{\text{annualized}} = \text{Standard Deviation}(\text{Daily Return}) \times \sqrt{252}$

Sortino Ratio

Focuses on downside volatility ($R_{p,t} < R_{f, \text{daily}}$) rather than total deviation:

Sortino Ratio Formula:

$$\text{Sortino Ratio} = (R_{\text{annualized}} - R_f) / \sigma_{\text{downside, annualized}}$$

Where $\sigma_{\text{downside, annualized}}$ is the standard deviation of only negative daily returns scaled by $\sqrt{252}$

CAPM Alpha (α) & Beta (β)

Beta (β): Measure of systemic market volatility matching the fund to its index (e.g. NIFTY 100):

CAPM Beta (β) Formula:

$$\text{Beta } (\beta) = \text{Covariance}(R_{\text{fund}}, R_{\text{market}}) / \text{Variance}(R_{\text{market}})$$

Alpha (α): Outperformance generated by the manager relative to the index adjusted for risk:

CAPM Alpha (α) Formula:

$$\text{Alpha } (\alpha) = R_{\text{fund, annualized}} - [R_f + \beta \times (R_{\text{market, annualized}} - R_f)]$$

Maximum Drawdown (MDD)

The peak-to-trough drop in NAV representing the worst-case potential loss:

Maximum Drawdown (MDD) Formulas:

$$\text{Drawdown}_t = (\text{NAV}_t - \text{Peak}_t) / \text{Peak}_t$$

$$\text{MDD} = \text{Minimum}(\text{Drawdown}_t)$$

6. Advanced Risk & Strategy Analytics

Beyond basic ratios, the analytics engine computes advanced risk models:

1. Value at Risk (VaR) & Conditional VaR (CVaR)

- **95% Daily VaR:** The 5th percentile of daily return history. It indicates that there is a 95% probability that the daily loss will not exceed this threshold.
- **95% Daily CVaR:** The expected value of daily losses given that they exceed the 95% VaR threshold (capturing tail-risk).

2. Herfindahl-Hirschman Index (HHI) for Sector Allocation

Calculates diversification concentration based on sector weights (w_i):

Herfindahl-Hirschman Index (HHI) Formula:

$$\text{HHI} = \sum (w_i)^2$$

Where w_i is the weight percentage of sector/stock i .

* **HHI < 1,500**: Diversified * **HHI 1,500 – 2,500**: Moderately Concentrated * **HHI > 2,500**: Highly Concentrated

3. Investor Cohort Analysis

Tracks groups of investors based on their onboarding quarter (cohort) to study long-term retention decay:

Retention Rate Formula:
 $\text{Retention Rate}_t = \text{Active Investors in Quarter } t / \text{Cohort Size}$

4. SIP Continuation & Retention Alerts

Classifies Systematic Investment Plan (SIP) investors by transaction recency:

- **Active:** Transaction observed within the last 35 days.
- **At Risk:** No transaction observed in the last 35 days (triggering retention efforts).

7. Business & Analytical Insights

Running analytical queries on the relational database yields structural insights:

A. Database Overview Metrics

- **Total Assets Under Management (AUM): 1,043,664.00 Crore INR** (~10.4 Lakh Crore INR)
- **Total Scheme Count: 40 Mutual Funds**
- **Total Transactions Audited: 32,778**
- **Active Investors: 5,000 Unique Customers**
- **Cumulative Transaction Value: 3,521,580,430.00 INR** (~352 Crore INR)

B. Mutual Fund Performance Scorecard (CAGR & Sharpe)

Funds are ranked based on a weighted scorecard (Sharpe: 30%, Alpha: 25%, CAGR: 20%, MDD: 15%, Expense Ratio: 10%):

[!NOTE] Mid-cap and Large-cap Equity funds dominated the top rankings due to substantial growth cycles in 2023-2025. Debt and Gilt funds remained stable but scored lower in composite lists due to lower growth.

Top 5 Scoring Funds:

1. **Mirae Asset Large Cap Fund - Regular - Growth** (CAGR: 14.81%, Sharpe: 1.760, MDD: -11.27%)
2. **Kotak Flexicap Fund - Regular - Growth** (CAGR: 15.65%, Sharpe: 1.568, MDD: -12.97%)
3. **ICICI Pru Midcap Fund - Regular - Growth** (CAGR: 18.08%, Sharpe: 1.391, MDD: -18.19%)
4. **Mirae Asset Tax Saver Fund - Regular - Growth** (CAGR: 14.95%, Sharpe: 1.468, MDD: -16.40%)
5. **HDFC Mid-Cap Opportunities Fund - Regular - Growth** (CAGR: 16.58%, Sharpe: 1.274, MDD: -16.22%)

Bottom 5 Scoring Funds (Performance or Drawdown Challenges):

1. **Axis Small Cap Fund - Regular - Growth** (CAGR: 1.52%, Sharpe: -0.179, MDD: -51.68%)

2. **UTI Mid Cap Fund - Regular - Growth** (CAGR: 1.17%, Sharpe: -0.266, MDD: -28.00%)
3. **SBI Small Cap Fund - Direct Plan - Growth** (CAGR: 2.05%, Sharpe: -0.158, MDD: -52.57%)
4. **HDFC Top 100 Fund - Regular Plan - Growth** (CAGR: 2.64%, Sharpe: -0.231, MDD: -24.73%)
5. **ABSL Small Cap Fund - Regular - Growth** (CAGR: 7.94%, Sharpe: 0.075, MDD: -35.45%)

C. Investor Demographics & Behavior

Age Group Distribution:

- **26-35 (Millennials): 1,451,600,218.00 INR** (41.2% market share by value)
- **36-45: 871,647,528.00 INR** (24.8% market share)
- **18-25 (Gen Z): 531,639,392.00 INR** (15.1% market share)
- **46-55: 405,406,469.00 INR** (11.5% market share)
- **56+ (Retirees): 261,286,823.00 INR** (7.4% market share)

Geographical Leaders:

- **Punjab:** 2,965 transactions totaling **315,780,459.00 INR**
- **Tamil Nadu:** 2,806 transactions totaling **315,177,237.00 INR**
- **Madhya Pradesh:** 2,931 transactions totaling **308,312,493.00 INR**

Payment Mode Penetration:

- Net Banking: **893.49M INR** (8,250 transactions)
- Cheque: **892.22M INR** (8,228 transactions)
- UPI: **888.24M INR** (8,154 transactions)
- Mandate (Auto-debit): **847.63M INR** (8,146 transactions)
- *Insight: The digital payment modes (UPI and Net Banking) have reached complete parity with traditional modes (Cheque and Mandate) in volume, reflecting a high financial digitization rate.*

D. Portfolio Concentration (HHI)

Out of 34 equity funds analyzed:

- **Diversified (< 1,500 HHI): 3 funds** (Average HHI: 1,342.56)
- **Moderately Concentrated (1,500 - 2,500 HHI): 27 funds** (Average HHI: 2,014.96)

Highly Concentrated (> 2,500 HHI): 4 funds (Average HHI: 2,641.02)

- *Insight: The majority of active equity schemes maintain a moderately concentrated position, focusing on key sectors like Banking (average weight 10.87%, total 62,840 Cr) and IT (average weight 11.39%, total 38,477 Cr).*

E. Cohort Retention Decay

Tracking quarterly customer cohorts reveals retention behaviors:

2024Q1 Cohort (Onboarded in Q1 2024): Started with **3,236 investors**.

- **Q1 (3 Months later):** **70.92%** active.
- **Q2 (6 Months later):** **70.49%** active.
- **Q4 (12 Months later):** **69.62%** active.
- *Insight: This cohort demonstrates highly stable retention, stabilizing near 69% after 1 year.*

2024Q2 Cohort: Started with **971 investors**.

- **Q1:** **63.13%** active.
- **Q2:** **63.85%** active.
- **Q4:** **48.71%** active.
- *Insight: Mid-year cohorts experience steeper drops, losing over half of their active base by Year 1.*

2024Q4 Cohort: Started with **186 investors**.

- **Q1:** **40.32%** active.
- **Q2:** **26.34%** active.
- *Insight: End-of-year cohorts display a critical churn risk, indicating high drop-offs. Strategic retention should target Q4 onboardings within their first 90 days.*

F. SIP Continuation Risk Summary

- **Active SIP Investors:** **1,241** (26.1%)
At Risk SIP Investors: **3,521** (73.9%)
- *Insight: Over 73% of SIP investors have not transacted in the last 35 days relative to the latest data date. This flags a systemic risk in continuous savings plans, suggesting a high number of paused or canceled SIP mandates.*

8. Interactive Presentation Layer (Dashboard)

The frontend application serves as the visual display for all database tables and analytics:

- **Page 1: Industry Overview:** Displays global KPIs (AUM, SIP Inflows, Folios, Schemes count) and charts detailing the growth of monthly SIP inflows.
- **Page 2: Fund Performance Scorecard:** Displays a searchable and filterable table of composite scores alongside a risk-return scatter chart (CAGR vs. Volatility).

- **Page 3: Drill-Through Analytics:** Renders a popup showing 5-day sampled historical NAV series vs. a rolling Sharpe ratio chart, together with sector holdings tables.
- **Page 4: Investor Demographics:** Visualizes state-by-state heatmaps, age-group pie charts, and payment mode breakdowns.
- **Page 5: Advanced Risk & Cohorts:** Renders heatmap tables for Cohort Retention, lists of at-risk SIP investors, and portfolio HHI concentration gauges.
- **Page 6: Recommendation Engine:** Formulates the top 3 mutual funds matching a user's chosen risk tolerance:

Chosen Risk Profile	Recommended Fund #1	Recommended Fund #2	Recommended Fund #3
Low Risk	ICICI Pru Liquid Fund - Regular (Sharpe: 2.506, CAGR: 7.68%)	Kotak Liquid Fund - Regular (Sharpe: 1.783, CAGR: 6.18%)	Mirae Asset Large Cap Fund - Regular (Sharpe: 1.760, CAGR: 14.81%)
Moderate Risk	Kotak Flexicap Fund - Regular (Sharpe: 1.568, CAGR: 15.65%)	Mirae Asset Emerging Bluechip Fund (Sharpe: 1.070, CAGR: 14.56%)	UTI Flexi Cap Fund - Regular (Sharpe: 0.688, CAGR: 15.34%)
High Risk	ICICI Pru Midcap Fund - Regular (Sharpe: 1.391, CAGR: 18.08%)	DSP Midcap Fund - Regular (Sharpe: 1.329, CAGR: 17.16%)	HDFC Mid-Cap Opportunities Fund (Sharpe: 1.274, CAGR: 16.58%)

9. Verification & Quality Assurance Results

All analytical tables and data pipeline stages have been validated by automated scripts:

```
>>> python verify_pipeline.py
### STARTING PIPELINE VERIFICATION ###
Verifying cleaned files in data/processed:
- clean_fund_master.csv: [PASS]
- clean_nav.csv: [PASS]
- clean_aum.csv: [PASS]
- clean_sip_inflows.csv: [PASS]
- clean_category_inflows.csv: [PASS]
- clean_folio_count.csv: [PASS]
- clean_performance.csv: [PASS]
- clean_transactions.csv: [PASS]
- clean_portfolio_holdings.csv: [PASS]
- clean_benchmark_indices.csv: [PASS]
[PASS] All 10 cleaned CSV files are present.

Verifying database bluestock_mf.db:
[PASS] bluestock_mf.db exists.
[PASS] All 10 database tables populated with exact row counts.
```


Granular data quality assertions:

```
[PASS] Assert: All NAV values in database are strictly greater than 0.  
[PASS] Assert: All transaction amounts in database are strictly greater than 0.  
[PASS] Assert: KYC status contains only expected values: ['Verified', 'Pending']  
[PASS] Assert: Transaction types contain only expected values: ['SIP', 'Redemption', 'Lumpsum']  
[PASS] Assert: Foreign key reference integrity checks clean.  
### PIPELINE VERIFICATION SUCCESSFUL: ALL CHECKS PASSED! ###
```

```
>>> python scripts/verify_analytics.py  
### STARTING ANALYTICS VERIFICATION PIPELINE ###  
Verifying fact_performance custom metrics:  
[PASS] All 40 funds have computed custom performance metrics.  
- Sharpe range: -0.395 to 2.506  
- Alpha range: -4.19% to 27.46%  
- Max Drawdown range: -52.57% to -0.10%  
  
Verifying fact_advanced_risk:  
[PASS] Advanced risk computed for exactly 40 schemes.  
[PASS] Assert: CVaR is strictly more negative or equal to VaR for all funds.  
  
Verifying fact_rolling_sharpe:  
[PASS] Found 10,600 rolling Sharpe records across 10 top funds.  
  
Verifying fact_cohort_metrics:  
[PASS] Found 21 cohort activity quarters.  
[PASS] Retention percentages lie within bounds: 24.1% to 100.0%  
  
Verifying fact_sip_continuation:  
[PASS] SIP continuation classification counts match (1,241 active vs 3,521 at risk).  
  
Verifying fact_sector_concentration:  
[PASS] HHI indices lie within standard mathematical bounds [100, 10000].  
  
Verifying fact_recommendations:  
[PASS] Engine outputted exactly the top 3 recommended schemes per risk profile.  
### ANALYTICS VERIFICATION SUCCESSFUL: ALL CHECKS PASSED! ###
```

10. Conclusion & Strategic Recommendations

The Mutual Fund Analytics pipeline is complete, verified, and serves a high-performance presentation layer. Based on the insights extracted, we suggest the following strategic initiatives:

1. **Millennial Engagement Focus:** Since investors aged 26-35 represent over 41% of transaction volume (1,451M INR), marketing and mobile features should target digital user experiences, particularly emphasizing fast UPI payment integrations.
2. **Q4 Cohort Retention Push:** Cohorts onboarded in Q4 show high drop-off rates (only 26.34% active by Q2). Trigger automated push notifications, educational newsletters, or loyalty incentives within the first 60 days of onboarding for Q4 users.
3. **Automated SIP Resumption Prompts:** With 73.9% of SIP mandates categorized as **At Risk**, the platform should implement automated triggers. When an investor skips a scheduled SIP contribution (no transaction for 30+ days), the system should suggest a one-click "Resume SIP" or "Pause for 3

Months" rather than letting the mandate lapse.

4. **HHI Concentration Alerts:** Educate investors holding "High Concentration" funds ($> 2,500$ HHI) on diversification. Recommend supplementary "Diversified" funds ($< 1,500$ HHI) from the recommendation database to lower their overall portfolio risk.